Additional Assessment 2MBA70

This assessment consists of 4 problems, reflecting results stated in the lecture notes, worth a total of 27 points. For solving them you are of course **not** allowed to simply use the corresponding result from the lecture notes. You are allowed to use any other results from the course and lecture notes and any results derived in this assessment.

Problem 1 (7 Points)
In this problem you will prove the result that states that it suffices to check measurability on the generating sets.
Let $(\Omega, \mathcal{F})$ and $(E, \mathcal{G})$ be two measurable spaces such that $\mathcal{G} = \sigma(\mathcal{A})$. Let $f : \Omega \to E$ be a function such that $f^{-1}(A) \in \mathcal{F}$ for all $A \in \mathcal{A}$.

(a) Consider the following collection of subsets:

$$\mathcal{H} := \{B \subset \mathcal{G} \,:\, f^{-1}(B) \in \mathcal{F}\}.$$

Show that $\mathcal{H}$ is a σ-algebra on E.

(b) Prove that f is $(\mathcal{F}, \mathcal{G})$-measurable. [Hint: use the inclusion property of generated σ-algebras to show that $\mathcal{G} \subseteq \mathcal{H}$.]

Problem 2 (6 points)
Let $(\Omega, \mathcal{F}, \mu)$ be a measure space, $f, g : \Omega \to \mathbb{R}_+$ two non-negative, measurable functions and $\alpha \geq 0$ be a constant.

(a) Suppose $B \in \mathcal{F}$ satisfies $\mu(B) = 0$. Prove that

$$\int_B f \,\mathrm{d}\mu = 0.$$

(b) Suppose $f \leq g$. Prove that

$$\int_\Omega f \,\mathrm{d}\mu \leq \int_\Omega g \,\mathrm{d}\mu.$$

(c) Prove that

$$\alpha \int_\Omega f \,\mathrm{d}\mu = \int_\Omega (\alpha f) \,\mathrm{d}\mu.$$

[General hint: use simple functions first.]
Problem 3 (5 Points)
Let $(X_n)_{n\geq 1}$ and X be random variables defined on the same probability space.
Prove that $X_n \xrightarrow{\mathbb{P}} X$ if and only if

$$\lim_{n\to\infty} \mathbb{P}(\|X_n - X\| > \varepsilon) = 0 \quad \text{for every } \varepsilon > 0.$$

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Problem 4 (9 Points)
Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $\mathcal{H}$ be a sub-σ-algebra of $\mathcal{F}$. Furthermore, let X, Y be random variables and $a \in \mathbb{R}$.

(a) Suppose that X, Y are $\mathcal{H}$-measureable and $\int_B X d\mathbb{P} = \int_B Y \mathbb{P}$ holds for all $B \in \mathcal{H}$. Prove that $X = Y$ $\mathbb{P}$-almost everywhere, i.e. $\int_\Omega |X - Y| d\mathbb{P} = 0$.

(b) If X is $\mathcal{H}$-measurable, prove that $\mathbb{E}[X|\mathcal{H}] = X$ holds $\mathbb{P}$-almost everywhere.

(c) $\mathbb{E}[X + Y|\mathcal{H}] = \mathbb{E}[X|\mathcal{H}] + \mathbb{E}[Y|\mathcal{H}]$ holds $\mathbb{P}$-almost everywhere.

(d) If $X \leq Y$, prove that $\mathbb{E}[X|\mathcal{H}] \leq \mathbb{E}[Y|\mathcal{H}]$ holds $\mathbb{P}$-almost everywhere.